



Ophthalmology Written Booklet — Clinoma Written



Essay Booklet – Written Questions (أسئلة مقالية)

✓ Answer Key Edition (ملحق الإجابات)
Total Chapters: 6 | Clinoma written

CURRICULUM

Ophthalmology Written

TOTAL CHAPTERS

6 Chapters

ACTIVE COURSE

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Chapter 1: الجفون (Eyelid)

Q 1

DEFINITION

Define blepharitis and enumerate its types.

Model Answer

Blepharitis is chronic inflammation of the eyelid margins.

Types include:

1. Ulcerative blepharitis: Bacterial infection (typically Staphylococcal), characterized by yellow crusts, glued lashes, and lash loss (madarosis).
2. Squamous/Scaly blepharitis: Seborrheic inflammation, characterized by greasy scales (dandruff-like) and lid hyperaemia without ulceration.
3. Allergic blepharitis: Hypersensitivity reaction (often to drugs or cosmetics), causing severe itching, scaling, and redness.

Q 2

DEFINITION

Define Blepharoptosis?

Model Answer

Blepharoptosis (ptosis) is an abnormal drooping of the upper eyelid, covering more than 2 mm of the upper part of the cornea in the primary gaze (normal lid covers only the upper 2 mm).

Q 3**ENUMERATE****Enumerate causes of acquired ptosis.****Model Answer**

1. Neurogenic: Due to 3rd nerve palsy (complete ptosis) or Horner's syndrome (sympathetic palsy leading to mild ptosis).
2. Myogenic: Due to muscular disorders like Myasthenia Gravis (characterized by fatigability and diurnal variation) or myotonic dystrophy.
3. Aponeurotic (Senile): Age-related dehiscence or disinsertion of the levator aponeurosis from the tarsal plate.
4. Mechanical: Due to excessive weight on the eyelid, such as orbital tumors, large chalazia, scarring, or severe inflammatory edema.

Q 4**SURGICAL PRINCIPLE****Give the surgical principle and choice of operation for ptosis correction based on levator function.****Model Answer**

The surgical principle is to raise the upper eyelid to match the normal eye by either shortening/advancing the levator muscle or suspending the lid to the frontalis muscle.

Choice of Operation:

1. If levator function is good or fair (≥ 5 mm excursion):
 - Levator resection / advancement (e.g., Blascovics internal approach or Everbusch external approach).
 - Fasanella-Servat operation (excision of tarsal plate, conjunctiva, and Müller's muscle) if ptosis is mild (1.5-2 mm) with good levator function.
2. If levator function is poor (≤ 4 mm excursion) with severe ptosis:
 - Frontalis suspension (sling) operation (e.g., Hess or Crawford operation using fascia lata or silicone sling) to connect the lid to the frontalis muscle.

Q 5**SURGICAL PRINCIPLE****What are the contraindications to ptosis surgery?****Model Answer**

1. Recent complete 3rd nerve palsy: Surgery is contraindicated until the squint is corrected to avoid intractable post-operative diplopia.
2. Absent Bell's phenomenon or neuroparalytic keratitis: High risk of severe post-operative exposure keratopathy and corneal ulceration.
3. Myasthenic ptosis: Medical treatment (Anticholinesterases/Systemic steroids) is primary; surgery is only for stable, medically refractory cases.
4. Corneal anesthesia: The drooped lid protects the anesthetic cornea; lifting it poses a high risk of exposure keratitis.

Q 6**DEFINITION****Define ectropion and name its types.****Model Answer**

Ectropion is an abnormal outward turning or eversion of the eyelid margin away from the globe.

Types:

1. Involutional (Senile): Age-related laxity of lid tissues (most common).
2. Cicatricial: Due to scarring of the skin/anterior lamella from trauma, burns, or chronic skin diseases.
3. Paralytic: Seventh (facial) nerve palsy causing orbicularis muscle weakness.
4. Spastic: Orbicularis contraction in a young patient with lid swelling.
5. Congenital: Rare development defect.

Q 7**SURGICAL PRINCIPLE****Give the surgical principle of correction for cicatricial ectropion.****Model Answer**

The surgical principle is to release the scarring (which shortens the anterior lamella) and restore the length of the anterior lamella of the eyelid.

This is achieved by scar excision/release, followed by skin lengthening procedures such as V-Y advancement, Z-plasty, or a full-thickness skin graft.

Q 8**SCENARIO**

A 65-year-old man presents with eversion of his left lower eyelid, epiphora, and exposure of the tarsal conjunctiva. (a) What is your diagnosis, and what are its degrees of severity? (b) Name the surgical principle for treating the most common type of this condition.

Model Answer

a) Diagnosis: Senile (Involutional) Ectropion of the lower eyelid.

Degrees of Severity:

- Mild: Eversion of the lacrimal punctum only (leads to epiphora).
- Moderate: Eversion of the lid margin with exposure of the tarsal conjunctiva.
- Severe: Complete eversion of the lid exposing the conjunctival fornix.

b) Surgical Principle for Senile Ectropion:

The principle is to tighten and shorten the horizontally lax lower lid and retighten the medial/lateral canthal tendons. Operations include the Kuhnt-Szymanowski procedure (horizontal shortening + skin excision) or the Lateral Tarsal Strip operation.

Q 9**DEFINITION****Define entropion and mention its etiological types.****Model Answer**

Entropion is the abnormal inward turning of the eyelid margin towards the globe, leading to rubbing of the eyelashes against the cornea (trichiasis).

Etiological Types:

1. Involutional (Senile): Age-related horizontal lid laxity and overriding of the preseptal orbicularis muscle over the pretarsal muscle.
2. Cicatricial: Scarring of the palpebral conjunctiva/posterior lamella (e.g., from Trachoma, chemical burns, Stevens-Johnson syndrome).
3. Spastic: Spasm of the orbicularis muscle secondary to ocular irritation or tight patching.
4. Congenital: Development defect of the tarsal plate.

Q 10**DEFINITION****Define Spastic entropion?****Model Answer**

Spastic entropion is the inward turning of the eyelid margin due to spasm of the marginal fibers of the orbicularis oculi muscle (muscle of Riolan), usually secondary to chronic ocular irritation, acute corneal/conjunctival inflammation, or tight bandaging, mostly occurring in elderly patients.

Q 11**ENUMERATE****Enumerate the lines of treatment for spastic entropion.****Model Answer**

1. Treat/remove the predisposing cause (e.g., treat ocular irritation, remove tight bandage).
2. Temporary measures: Everting sutures, Botulinum toxin (Botox) injection to paralyze the orbicularis muscle, or pulling the lid down with a T-shaped adhesive plaster.
3. Permanent surgical measures (for chronic/recurrent cases): Wheeler's skin-and-muscle operation (resection of an elliptical strip of skin and underlying muscle of Riolan) to correct the overriding orbicularis.

Q 12**DEFINITION****Define lagophthalmos and enumerate its local causes.****Model Answer**

Lagophthalmos is the incomplete closure of the palpebral fissure when voluntary or involuntary closure of the eyelids is attempted.

Local Causes:

1. Lid lesions: Severe ectropion, structural loss/coloboma, or overcorrection during ptosis surgery.
2. Mechanical/Spastic: Eyelid retraction (e.g., Thyroid eye disease / Dalrymple's sign).
3. Paralytic: Lower motor neuron facial (7th) nerve palsy (Bell's palsy) paralyzing the orbicularis oculi muscle.
4. Globe lesions: Severe proptosis, anterior staphyloma, or buphthalmos.

Q 13**ENUMERATE****Enumerate complications of lagophthalmos.****Model Answer**

1. Exposure keratopathy (dryness of the lower third of the cornea).
2. Corneal ulceration (dryness leading to epithelial desquamation and secondary bacterial infection).
3. Corneal perforation and endophthalmitis in severe untreated cases.
4. Epiphora due to loss of the lacrimal pump mechanism.

Q 14**DEFINITION****Define trichiasis, enumerate its causes, and differentiate it from distichiasis.****Model Answer**

Trichiasis is an acquired condition in which the eyelashes are directed posteriorly, rubbing against the cornea and conjunctiva, arising from normal follicles.

Causes:

- Trachoma (most common in endemic regions).
- Ulcerative blepharitis.
- Eyelid trauma or chemical/thermal burns.
- Cicatrizing conjunctivitis (Stevens-Johnson syndrome, Ocular Cicatricial Pemphigoid).

Difference from Distichiasis:

- Distichiasis is a congenital condition where an extra row of eyelashes arises from the Meibomian gland orifices behind the grey line.
- Trichiasis is an acquired misdirection of normal lashes from normal follicles.

Q 15**DEFINITION****Define Hordeolum externum (Stye)?****Model Answer**

Hordeolum externum (Stye) is an acute, painful, purulent bacterial infection (typically *Staphylococcus aureus*) of the eyelash follicle and its associated glands of Zeis or Moll.

Chapter 2: القرنية (Cornea)

Q 1**DEFINITION****Define keratitis and enumerate its main types.****Model Answer**

Keratitis is the inflammation of the cornea, characterized by ciliary congestion, corneal infiltration, and loss of transparency.

Main Types:

1. Bacterial Keratitis: Rapidly progressive, causing purulent hypopyon ulcers (e.g., *Pneumococcus*, *Pseudomonas*).
2. Viral Keratitis: Dendritic or geographic ulceration (*Herpes Simplex Virus*), or disciform stromal keratitis.
3. Fungal Keratitis: Dry, fluffy, feathery margins with satellite lesions, common in trauma with vegetative matter (e.g., *Aspergillus*, *Candida*).
4. Acanthamoeba Keratitis: Associated with contact lens wear and exposure to contaminated water, presenting with severe out-of-proportion pain and ring-shaped infiltrates.

Q 2**COMPARISON****Compare between a hypopyon ulcer and a dendritic ulcer.****Model Answer**

Feature	Hypopyon Ulcer	Dendritic Ulcer
Causative Organism	Bacteria (Pneumococcus, Pseudomonas)	Herpes Simplex Virus (HSV-1)
Pain	Severe pain	Mild pain (due to corneal anesthesia)
Shape	Circular, crescentic, or serpiginous with yellow margins	Linear, branching tree-like with terminal knobs
Corneal Sensation	Normal	Decreased or anesthetic
Hypopyon	Common, sterile, and early	Rare (unless secondary bacterial infection)
Treatment	Intensive topical antibiotics, cycloplegics	Topical antivirals (Acyclovir), STEROIDS CONTRAINDICATED

Q 3**SCENARIO**

A 30-year-old contact lens wearer presents with a painful red eye, photophobia, and a corneal ulcer with a grey, dry, fluffy surface and feathery margins. (a) What is the most likely diagnosis and causative organism? (b) How do you confirm the diagnosis? (c) Enumerate the lines of treatment.

Model Answer

a) Diagnosis: Fungal Keratitis (Keratomycosis).

Causative Organisms: Filamentous fungi (*Aspergillus* or *Fusarium* species) or yeasts (*Candida*).

b) Confirmation of Diagnosis:

- Corneal scrapings from the base and edges of the ulcer.
- Staining: Gram and Giemsa stains (reveal hyphae) or KOH wet mount.
- Culture: Inoculation on Sabouraud's dextrose agar (SDA) incubated at room temperature.

c) Lines of Treatment:

1. Medical:

- Topical antifungal drops: Natamycin 5% (for filamentous fungi) or Amphotericin B 0.15% (for *Candida*).
- Systemic antifungals (e.g., Voriconazole, Itraconazole) for deep stromal involvement.
- Cycloplegic drops (Atropine 1%) to relieve ciliary spasm and prevent synechiae.
- Avoid corticosteroids.

2. Surgical (if medical therapy fails):

- Debridement (therapeutic scraping).
- Therapeutic Penetrating Keratoplasty (PKP) to restore integrity.

Q 4**ENUMERATE**

Enumerate causes of corneal perforation.

Model Answer

1. Untreated or severe bacterial corneal ulcers (particularly Pneumococcal or *Pseudomonas*).
2. Fungal keratitis with deep stromal necrosis.
3. Chemical injuries (especially alkaline burns causing liquefactive necrosis).
4. Keratomalacia (severe Vitamin A deficiency causing melting of the cornea).
5. Mechanical trauma (penetrating corneal injury).

Q 5**SURGICAL PRINCIPLE****Give the surgical principle of management for corneal perforation.****Model Answer**

The surgical principle is to urgently seal the perforation to restore the integrity of the globe, reform the anterior chamber, and prevent endophthalmitis.

Management Options:

- Small perforations (< 1 mm): Application of cyanoacrylate tissue glue and a bandage contact lens (BCL).
- Medium perforations: Patch grafting (using amniotic membrane, cornea, or sclera).
- Large perforations (> 2 mm): Urgent therapeutic penetrating keratoplasty (PKP).

Q 6**SURGICAL PRINCIPLE****Explain the surgical principle of treating a corneal fistula and mention the options if it fails.****Model Answer**

The surgical principle is to destroy the epithelial lining that has grown to line the tract of the fistula, allowing the tract to close and heal by fibrosis.

Method:

- Chauffage: Indirect thermal cauterization close to the fistula opening.

Options if it fails:

- Therapeutic penetrating keratoplasty (PKP) to replace the fistulized area with a healthy donor button.

Q 7**ENUMERATE****Enumerate the complications of a non-perforated corneal ulcer.****Model Answer**

1. Secondary Iridocyclitis: Due to toxins diffusion into the anterior chamber, presenting with hypopyon.
2. Secondary Glaucoma (Inflammatory): Blockage of trabecular meshwork by inflammatory cells/exudates.
3. Descemetocoele: Herniation of Descemet's membrane due to destruction of the stroma.
4. Corneal Opacities: Healing by scar tissue forming Nebula (faint), Macula (semi-dense), or Leucoma (dense white opacity).
5. Ectatic Scar (Corneal Ectasia): Thinning of the cornea causing it to bulge forward under normal IOP.

Q 8**DEFINITION****Define keratoconus and name its signs and lines of management.****Model Answer**

Keratoconus is a bilateral, non-inflammatory, progressive developmental ectatic dystrophy in which the central or paracentral stroma thins, causing the cornea to bulge forward in a conical shape.

Signs:

- Irregular astigmatism with scissoring reflex on retinoscopy.
- Vogt striae (fine vertical stress lines in deep stroma).
- Fleischer ring (iron deposition in epithelium at the base of the cone).
- Munson sign (bulging of the lower lid on down-gaze).
- Rizzuti's sign (conical light reflex on nasal limbus when light is shone from temporal side).

Management:

1. Spectacles or soft contact lenses (early cases for mild astigmatism).
2. Rigid Gas Permeable (RGP) contact lenses (to mask irregular astigmatism).
3. Corneal Collagen Cross-linking (CXL) with Riboflavin and UV-A (to stop progression).
4. Intrastromal Corneal Ring Segments (ICRS) (to flatten the cone and improve lens tolerance).
5. Penetrating or Deep Anterior Lamellar Keratoplasty (DALK) for advanced cases with apical scarring.

Q 9**DEFINITION****Define keratoplasty and enumerate its types and indications.****Model Answer**

Keratoplasty (corneal graft) is the surgical replacement of diseased or scarred host corneal tissue with healthy donor corneal tissue.

Types:

1. Penetrating Keratoplasty (PKP): Full-thickness replacement.
2. Lamellar Keratoplasty: Partial-thickness replacement, e.g., Deep Anterior Lamellar Keratoplasty (DALK) or Descemet's Stripping Automated Endothelial Keratoplasty (DSAEK).

Indications:

1. Optical: To improve visual acuity by replacing an opaque or scarred cornea (e.g., post-ulcer scar, keratoconus, corneal dystrophy).
2. Tectonic: To restore structural integrity of a thin or perforated cornea (e.g., descemetocoele, corneal perforation).
3. Therapeutic: To remove active resistant infectious tissue (e.g., fungal ulcer unresponsive to medical treatment).
4. Cosmetic: To improve appearance in a blind eye.

Q 10**ENUMERATE****Enumerate the layers of the Cornea from anterior to posterior.****Model Answer**

1. Epithelium (stratified squamous non-keratinized).
2. Bowman's membrane (acellular collagenous layer, does not regenerate).
3. Stroma (constitutes 90% of thickness, regular arrangement of collagen fibrils).
4. Descemet's membrane (basement membrane of endothelium, regenerates).
5. Endothelium (single layer of hexagonal cells, maintains dehydration, does not regenerate).

 Chapter 3: الشبكية (Retina)**Q 1****DEFINITION****Define retinal detachment and enumerate its types.****Model Answer**

Retinal detachment is the separation of the neurosensory retina (photoreceptor layer) from the underlying retinal pigment epithelium (RPE).

Types:

1. Rhegmatogenous: Caused by a full-thickness break (hole/tear) allowing liquefied vitreous into the subretinal space.
2. Tractional: Caused by contraction of fibrovascular membranes pulling the retina (e.g., in proliferative diabetic retinopathy or ROP).
3. Exudative (Serous): Caused by accumulation of subretinal fluid in the absence of a break, due to inflammation, neoplasm (e.g., choroidal melanoma), or systemic hypertension.

Q 2**DEFINITION****Define rhegmatogenous retinal detachment and enumerate its risk factors.****Model Answer**

Rhegmatogenous retinal detachment is a type of retinal detachment initiated by a break (hole or tear) in the retina, which allows fluid from the liquefied vitreous cavity to enter the subretinal space.

Risk Factors:

1. High myopia (due to peripheral retinal thinning/degeneration like lattice degeneration).
2. Previous intraocular surgery (especially cataract surgery with vitreous loss).
3. Trauma to the globe.
4. Retinal detachment in the fellow eye.
5. Family history of retinal detachment.

Q 3**SURGICAL PRINCIPLE**

Give the surgical principle of scleral buckling for rhegmatogenous retinal detachment.

Model Answer

The surgical principle is to close the retinal break and relieve vitreous traction.

Steps:

1. Localization of the retinal break.
2. Cryotherapy or laser photocoagulation around the break to create a chorioretinal scar that seals the hole.
3. Indentation of the sclera (scleral buckling) using a silicone sponge or band (explant) to press the RPE against the detached retina.
4. Subretinal fluid drainage (optional) to allow immediate reapposition.

Q 4**ENUMERATE**

Enumerate signs of proliferative diabetic retinopathy.

Model Answer

1. Neovascularization at the optic disc (NVD) or neovascularization elsewhere (NVE) on the retina.
2. Fibrovascular membranes growing on the surface of the retina.
3. Preretinal (subhyaloid) hemorrhage or vitreous hemorrhage.
4. Tractional retinal detachment due to contracting fibrovascular bands.

Q 5**ENUMERATE**

Enumerate the signs of pre-proliferative diabetic retinopathy.

Model Answer

1. Multiple cotton wool spots (soft exudates, representing microinfarcts of nerve fiber layer).
2. Venous changes (venous beading, loops, or sausage-like dilation).
3. Intraretinal microvascular abnormalities (IRMA, representing dilated capillary shunts).
4. Extensive intraretinal blot hemorrhages.

Q 6**ENUMERATE**

Enumerate the indications for laser photocoagulation in diabetic retinopathy.

Model Answer

1. Panretinal Photocoagulation (PRP): Indicated in Proliferative Diabetic Retinopathy (PDR) or very severe Non-Proliferative Diabetic Retinopathy (NPDR) with high risk of progression.
2. Focal or Grid Laser: Indicated in Clinically Significant Macular Edema (CSME) to reduce vascular leakage at the macula.

Q 7**SCENARIO**

A 60-year-old hypertensive diabetic patient presents with sudden painless loss of vision in his right eye. Fundus examination reveals extensive intraretinal hemorrhages, cotton wool spots, and disc edema. (a) What is the most probable diagnosis? (b) How do you differentiate between the ischemic and non-ischemic types?

Model Answer

a) Diagnosis: Central Retinal Vein Occlusion (CRVO) (often termed "blood and thunder" fundus).

b) Differentiator Table:

Feature | Non-Ischemic CRVO | Ischemic CRVO

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Visual Acuity | Mild-to-moderate impairment (usually > 6/60) | Severe impairment (usually < 6/60 or PL)

Afferent Pupillary Defect (RAPD) | Absent or very mild | Marked (positive RAPD)

Fundus hemorrhages | Moderate hemorrhages | Extensive, dense "splash-of-tomato-juice" hemorrhages

Cotton wool spots | Few or absent | Numerous (> 10)

Fluorescein Angiography | Small areas of capillary non-perfusion | Large areas (> 10 disc areas) of capillary non-perfusion

Complications | Good prognosis; low rate of rubeosis iridis | High risk (50%) of developing rubeosis iridis and neovascular glaucoma ("100-day glaucoma")

Q 8**ENUMERATE**

Name causes of cherry-red spot.

Model Answer

1. Central Retinal Artery Occlusion (CRAO - due to opaque edematous macula contrasting with the vascular choroid at the foveola).
2. Lipid storage diseases: Tay-Sachs disease, Niemann-Pick disease.
3. Trauma (Berlin's edema/commotio retinae at the macula).
4. Quinine toxicity.

Q 9**SURGICAL PRINCIPLE**

What is the surgical principle of treating central retinal artery occlusion (CRAO) and name three emergency measures.

Model Answer

The principle is to dislodge the embolus from the main trunk of the central retinal artery into a more peripheral branch, and to increase retinal perfusion by lowering intraocular pressure (IOP) and dilating the retinal vessels.

Emergency Measures (Must be done within 90-120 minutes):

1. Firm ocular massage (to repeatedly compress and release the globe to dislodge the embolus).
2. Rapid lowering of IOP: Intravenous or oral Acetazolamide (Diamox), intravenous Mannitol 20%, or anterior chamber paracentesis.
3. Vasodilation: Sublingual Nitroglycerin or inhalation of Carbogen (95% O₂ and 5% CO₂) to dilate the central retinal artery and increase oxygenation.

Q 10**DEFINITION**

Define retinitis pigmentosa and describe the classic triad of signs.

Model Answer

Retinitis pigmentosa (pigmentary retinal dystrophy) is a bilateral, genetically determined, progressive dystrophy of the photoreceptors (initially rods, then cones) and RPE.

Classic Triad of Signs:

1. Bone-spicule pigment deposits (clumping of pigment in the mid-peripheral retina, starting along vessels).
2. Retinal arteriolar attenuation (narrowing of the retinal blood vessels).
3. Waxy disc pallor (yellowish-white optic disc due to glial overgrowth).

Q 11**ENUMERATE**

Enumerate optic nerve head changes in open-angle glaucoma.

Model Answer

1. Progressive enlargement and deepening of the optic cup (increased Cup/Disc ratio > 0.6 or asymmetry > 0.2 between eyes).
2. Thinning and notching of the neuroretinal rim (especially at the superior and inferior poles - ISNT rule violation).
3. Nasalization of retinal vessels (shifting of vessels towards the nasal side of the disc).
4. Bayonetting sign (vessels making sharp double bends as they cross the excavation margin).
5. Splinter hemorrhages (Drance hemorrhages) at the disc margin.

Q 1**DEFINITION**

Define myopia, describe its etiological types, and the fundus changes in high myopia.

Model Answer

Myopia (nearsightedness) is a refractive state of the eye in which parallel rays of light focus in front of the retina when accommodation is relaxed.

Etiological Types:

1. Axial Myopia: Increased anteroposterior axial length of the eyeball (most common).
2. Curvature Myopia: Increased curvature of the cornea or lens (e.g., keratoconus, anterior lenticonus).
3. Index Myopia: Increased refractive index of the lens (e.g., nuclear sclerosis in senile cataract).
4. Positional Myopia: Anterior displacement of the lens (e.g., post-trauma).

Fundus Changes in High Myopia (Pathological Myopia):

- Tessellated/Tigroid fundus (prominent choroidal vessels).
- Myopic crescent (crescentic white patch at temporal border of disc).
- Posterior staphyloma (outpouching of sclera at posterior pole).
- Fuchs' spot (macular hemorrhage leading to subretinal neovascularization/scarring).
- Lattice degeneration, cobweb vitreous floaters, and risk of retinal breaks/detachment.

Q 2**ENUMERATE**

Enumerate types of astigmatism.

Model Answer

Astigmatism is classified into:

1. Regular Astigmatism (principal meridians are perpendicular to each other):
 - Simple Myopic: One meridian is myopic, other is emmetropic.
 - Simple Hypermetropic: One meridian is hypermetropic, other is emmetropic.
 - Compound Myopic: Both meridians are myopic (different degrees).
 - Compound Hypermetropic: Both meridians are hypermetropic (different degrees).
 - Mixed: One meridian is myopic, other is hypermetropic.
2. Irregular Astigmatism: Principal meridians are not perpendicular, and refractive power varies in different parts of same meridian (e.g., due to corneal scarring or keratoconus).

Q 3**DEFINITION****Define astigmatism and describe its types according to the rule.****Model Answer**

Astigmatism is a refractive error where the refractive power of the eye is unequal in different meridians, so light rays cannot focus to a single point on the retina.

Types according to the rule:

1. With-the-rule Astigmatism: The vertical meridian has the greatest refractive power (is most curved) (corrected by a minus cylinder at 180° or plus cylinder at 90°).
2. Against-the-rule Astigmatism: The horizontal meridian has the greatest refractive power (is most curved) (corrected by a minus cylinder at 90° or plus cylinder at 180°).
3. Oblique Astigmatism: Principal meridians lie between 30° and 60° or 120° and 150°.

Q 4**COMPARISON****Differentiate between simple, compound, and mixed astigmatism with examples.****Model Answer**

Type	Definition	Example Prescription
Simple Astigmatism	One principal meridian is emmetropic (focuses on retina), and the other is ametropic (focuses in front of or behind retina).	Plano / -2.00 DS x 180° (Simple Myopic)
Compound Astigmatism	Both principal meridians are ametropic of the same type (either both myopic or both hypermetropic) but of different degrees.	-1.50 DS / -1.00 DC x 180° (Compound Myopic)
Mixed Astigmatism	One principal meridian is myopic (focuses in front of retina), and the other is hypermetropic (focuses behind retina).	+2.00 DS / -3.50 DC x 180° (Mixed Astigmatism)

Q 5**DEFINITION****Define hypermetropia, describe its components and complications.****Model Answer**

Hypermetropia (farsightedness) is a refractive state of the eye in which parallel rays of light focus behind the retina when accommodation is relaxed.

Components:

1. Total Hypermetropia: The entire refractive error under complete cycloplegia.
2. Latent Hypermetropia: Corrected by the normal tonic contraction of the ciliary muscle (~1.0 Diopter).
3. Manifest Hypermetropia: Total minus latent, which consists of:
 - Facultative: Can be overcome by voluntary accommodation.
 - Absolute: Cannot be overcome by accommodation (requires glasses for distance).

Complications:

- Accommodative asthenopia (eyestrain, headache on near work).
- Accommodative esotropia (convergent squint in children due to high accommodative-convergence demand).
- Amblyopia (lazy eye) if unilateral or high bilateral hypermetropia goes uncorrected.
- Predisposition to primary angle-closure glaucoma (due to crowded anterior segment).

Q 6**SURGICAL PRINCIPLE****Give the principle of correction for unilateral aphakia.****Model Answer**

Unilateral aphakia (loss of lens in one eye) creates severe aniseikonia (difference in image size between eyes, ~30%) if corrected with spectacles, preventing binocular single vision.

Principle of Correction:

1. Secondary Intraocular Lens (IOL) implantation: Ideal method, placing the lens close to the nodal point of the eye to achieve normal image size (~1% aniseikonia).
2. Contact lenses: Suitable alternative if surgery is contraindicated (~5-7% aniseikonia, which is tolerated by the brain).
3. Refractive surgery (corneal inlay/epikeratophakia): Rarely used.

Q 7**CALCULATION**

An emmetropic patient at 45 years of age finds it difficult to read at 33 cm. His near point (Punctum Proximum) is 50 cm. Calculate the power of the reading glasses he needs.

Model Answer

1. Calculate Amplitude of Accommodation (AoA):

$AoA = 1 / \text{Near Point (in meters)} = 1 / 0.5 \text{ m} = 2.0 \text{ Diopters}$.

2. Determine comfortable accommodation range:

A patient can comfortably use only 2/3 of their total amplitude of accommodation, reserving 1/3 in reserve.

Comfortable (allowable) accommodation = $2/3 * 2.0 \text{ D} = 1.33 \text{ Diopters}$.

3. Calculate total power required for reading at 33 cm:

Required Power = $1 / \text{Reading Distance (in meters)} = 1 / 0.33 \text{ m} = 3.0 \text{ Diopters}$.

4. Calculate the reading glasses prescription:

Reading Addition = Required Power - Comfortable Accommodation = $3.0 \text{ D} - 1.33 \text{ D} = +1.67 \text{ Diopters}$.

Prescription: The patient needs reading glasses of approximately +1.75 D (or +1.50 D).

Q 8**SURGICAL PRINCIPLE**

Name two suitable corneal refractive surgical options and list three absolute contraindications.

Model Answer

Corneal Refractive Surgical Options:

1. LASIK (Laser-Assisted In Situ Keratomileusis): Flap created in corneal stroma, and excimer laser ablation performed underneath.
2. PRK (Photorefractive Keratectomy) / Trans-PRK: Surface ablation of the corneal stroma after epithelial removal.
3. SMILE (Small Incision Lenticule Extraction): Femtosecond laser creates and extracts a stromal lenticule through a microscopic incision.

Absolute Contraindications:

1. Keratoconus, suspected keratoconus, or abnormal corneal topography.
2. Inadequate corneal thickness (calculated residual stromal bed thickness under flap < 250 microns).
3. Active ocular infection or severe ocular surface inflammation (e.g., severe dry eye, active uveitis).
4. Severe systemic connective tissue disease (e.g., systemic lupus erythematosus, rheumatoid arthritis).

Q 9**SURGICAL PRINCIPLE**

Explain the optical principle of the 'Pinhole test' and what it indicates if vision improves with it.

Model Answer**Optical Principle:**

The pinhole acts as an artificial pupil that blocks peripheral divergent light rays (which cause spherical aberration and blur in refractive errors) and only allows central, parallel light rays to enter the eye and reach the fovea directly.

Clinical Indication:

- If visual acuity improves with the pinhole: The visual loss is primarily due to an uncorrected error of refraction or corneal irregularity.
- If visual acuity does not improve: The visual loss is due to an organic pathology (such as macular disease, vitreous opacity, optic nerve lesion, or advanced cataract).

 Chapter 5: الملتهمة (Conjunctiva)**Q 1****DEFINITION**

Define conjunctivitis and enumerate its causes.

Model Answer

Conjunctivitis is the inflammation of the conjunctiva, clinically presenting with conjunctival congestion (red eye), discharge, and foreign body sensation.

Causes:

1. Bacterial: Acute purulent (e.g., Pneumococcal), hyperacute (Gonococcal), chronic (associated with blepharitis).
2. Chlamydial: Trachoma, neonatal inclusion conjunctivitis.
3. Viral: Adenovirus (epidemic keratoconjunctivitis), Herpes Simplex Virus.
4. Allergic: Vernal keratoconjunctivitis (spring catarrh), giant papillary conjunctivitis.
5. Chemical/Toxic: Acid/alkali splash, toxic drugs.

Q 2**ENUMERATE****Enumerate signs of membranous conjunctivitis.****Model Answer**

Membranous conjunctivitis is a severe inflammation characterized by coagulation of fibrinous exudate on the conjunctival surface.

Signs:

1. True greyish-white membrane covering the palpebral conjunctiva.
2. Bleeding of the underlying conjunctival surface when attempting to peel the membrane.
3. Marked swelling and induration of the eyelids.
4. Preauricular lymphadenopathy.
5. Associated corneal complications (ulceration/perforation).

Q 3**SURGICAL PRINCIPLE****Give the principle of treatment for papillary conjunctivitis.****Model Answer**

The principle of treatment is to identify and remove the causal mechanical or allergic stimulus, and suppress the inflammatory/allergic pathway.

Treatment Steps:

1. Eliminate the trigger: Stop contact lens wear, remove prosthesis, or avoid allergens.
2. Pharmacotherapy: Mast cell stabilizers (e.g., Sodium cromoglycate) and antihistamines.
3. Anti-inflammatories: Mild topical steroids (e.g., fluorometholone) in severe cases under strict monitoring.
4. Lubricants: Preservative-free artificial tears to wash away mediators.

Q 4**ENUMERATE****Name causes of chronic conjunctivitis.****Model Answer**

1. Untreated or chronic bacterial blepharoconjunctivitis (Staphylococcal).
2. Chronic Chlamydial infection (Trachoma).
3. Persistent allergic exposure (allergic conjunctivitis).
4. Chronic dacryocystitis (reservoir of infection).
5. Ocular surface dryness (severe dry eye/Sjögren's syndrome).
6. Chronic exposure to chemical irritants, dust, or smoke.

Q 5**DEFINITION****Define trachoma and enumerate its eyelid and corneal complications.****Model Answer**

Trachoma is a chronic keratoconjunctivitis caused by *Chlamydia trachomatis* (serotypes A, B, Ba, and C), endemic in developing areas, characterized by follicles, papillae, and pannus, healing with cicatrization.

Eyelid Complications:

- Trichiasis (misdirected lashes).
- Cicatricial Entropion (inward lid rotation due to conjunctival scarring - Arlt's line).
- Ptosis (due to scarring of Müller's muscle and lid thickening - tylosis).
- Madarosis or Ankyloblepharon.

Corneal Complications:

- Trachomatous corneal ulcer (due to mechanical irritation of trichiasis).
- Corneal opacities (nebula/leucoma) leading to blindness.
- Pannus (superficial corneal vascularization and infiltration).
- Xerosis (severe dryness due to destruction of goblet cells and lacrimal ducts).

Q 6**ENUMERATE****Enumerate the clinical types and manifestations of Vernal Kerato-Conjunctivitis (Spring Catarrh).****Model Answer**

Vernal Kerato-conjunctivitis (VKC) is a bilateral, recurrent, seasonal allergic inflammation, predominantly affecting young boys.

Clinical Types:

1. Palpebral Type: Giant cobblestone papillae on the upper tarsal conjunctiva.
2. Bulbar/Limbal Type: Gelatinous thickening at the limbus with white Horner-Trantas dots (clumps of eosinophils).
3. Mixed Type: Combination of palpebral and bulbar features.

Ocular/Corneal Manifestations:

- Severe itching, photophobia, and ropy mucoid discharge.
- Shield ulcer (sterile transverse ulcer in upper cornea due to friction of giant papillae).
- Punctate epithelial keratitis.

Q 7**SCENARIO**

A 10-year-old boy presents with severe bilateral itching, ropy mucoid discharge, and photophobia. On examination, you see large, flat-topped papillae on the upper tarsal conjunctiva. (a) What is your diagnosis? (b) How do you differentiate these papillae from trachomatous papillae?

Model Answer

a) Diagnosis: Vernal Kerato-conjunctivitis (VKC) - Palpebral type.

b) Differentiator Table:

Feature | VKC Papillae | Trachomatous Papillae

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Etiology | IgE-mediated allergic reaction | Chronic infection (Chlamydia trachomatis)

Size | Giant ("Cobblestone" appearance, > 1 mm) | Fine, small, velvety red spots

Color | Bluish-white (milky hue due to mucus/fibrosis) | Bright red (hyperemic)

Upper Fornix | Free from lesions | Scarred and involved (follicles present)

Discharge | Thick, ropy, elastic mucoid discharge | Mucopurulent discharge (rich in bacteria, no eosinophils)

Q 8**DEFINITION**

Define pannus and describe its clinical types.

Model Answer

Pannus is the superficial infiltration of the cornea with inflammatory cells and newly formed blood vessels from the limbus into the upper part of the cornea (between Bowman's membrane and epithelium).

Clinical Types:

1. Pannus Tenuis: Thin, early pannus with few vessels and mild infiltration.
2. Pannus Vasculosus: Highly vascularized, active pannus with multiple engorged vessels.
3. Pannus Carnosus (Fleshy): Thick, red, gelatinous pannus indicating severe active disease.
4. Pannus Annularis: Rare form forming a complete vascular ring around the cornea.
5. Pannus Siccus (Cicatrized): Inactive, healed pannus characterized by white fibrous tissue and empty "ghost" vessels.

Q 9**DEFINITION**

Define Ophthalmia Neonatorum and name the causative organisms and their time of onset.

Model Answer

Ophthalmia Neonatorum is any acute conjunctivitis occurring in a newborn during the first month of life (acquired during passage through the birth canal).

Causes & Time of Onset:

1. Chemical (e.g., Silver nitrate drops prophylaxis): Appears within the first 24 hours (resolves spontaneously).
2. Gonococcal (*Neisseria gonorrhoeae*): Appears 2 to 5 days after birth (hyperacute purulent conjunctivitis, risk of rapid corneal perforation).
3. Chlamydial (*Chlamydia trachomatis* - Inclusion blennorrhea): Appears 5 to 14 days after birth (mucopurulent, most common infectious cause).
4. Viral (Herpes Simplex Virus-2): Appears 1 to 2 weeks after birth (vesicles on lids, geographic keratitis).
5. Other Bacteria (*Staphylococcus*, *Streptococcus*): Appears 5 to 10 days after birth.

Q 10**SURGICAL PRINCIPLE**

What is the surgical principle for treating a recurrent pterygium?

Model Answer

The surgical principle is to excise the pterygium head and body completely, and then create a barrier to prevent the migration of subconjunctival fibroblasts and vessels across the limbus.

Methods to prevent recurrence:

1. Wide excision with conjunctival autograft (CAG) (harvested from superior bulbar conjunctiva of same eye) secured with sutures or fibrin glue.
2. Excision with amniotic membrane graft (AMG).
3. Intraoperative application of Mitomycin C (MMC 0.02% for 2-3 minutes) to inhibit fibrovascular proliferation.
4. Post-operative Beta irradiation (Strontium-90).

Q 1**DEFINITION****Define proptosis and enumerate its common causes.****Model Answer**

Proptosis (exophthalmos) is the abnormal protrusion or forward displacement of the eyeball from the orbit.

Common Causes:

1. Endocrine: Thyroid eye disease (Graves' ophthalmopathy - most common cause of bilateral and unilateral proptosis).
2. Inflammatory: Orbital cellulitis, orbital pseudotumor (nonspecific orbital inflammation).
3. Vascular: Carotid-cavernous sinus fistula (pulsating proptosis), orbital varices.
4. Neoplastic:
 - Benign: Cavernous hemangioma, optic nerve glioma.
 - Malignant: Rhabdomyosarcoma (children), lacrimal gland carcinoma, orbital metastasis.
5. Traumatic: Retrobulbar hemorrhage, orbital roof fracture.

Q 2**DEFINITION****Define orbital cellulitis and enumerate its potential complications.****Model Answer**

Orbital cellulitis is an acute, purulent, life- and vision-threatening infection of the soft tissues of the orbit posterior to the orbital septum, usually secondary to paranasal sinusitis (especially ethmoidal).

Complications:

1. Ocular:
 - Exposure keratopathy and corneal ulceration.
 - Compressive optic neuropathy (leading to permanent blindness).
 - Central retinal artery occlusion (CRAO) or central retinal vein occlusion (CRVO).
 - Orbital abscess or subperiosteal abscess.
 - Panophthalmitis.
2. Intracranial:
 - Cavernous sinus thrombosis (life-threatening bilateral ophthalmoplegia and sepsis).
 - Meningitis and extradural/brain abscess.
3. Systemic: Septicemia and pyaemia.

Q 3**SURGICAL PRINCIPLE****Give the principle of management for orbital cellulitis.****Model Answer**

The principle of management is urgent hospitalization and aggressive treatment to control infection, protect the optic nerve, and prevent intracranial spread.

Management Steps:

1. Admission: Urgent hospitalization.
2. Systemic Antibiotics: High-dose intravenous broad-spectrum antibiotics (e.g., Ceftriaxone + Vancomycin or Metronidazole) adjusted after cultures.
3. Monitoring: Frequent checks of visual acuity, pupillary reflexes, and extraocular motility.
4. Imaging: Urgent CT scan of orbit and brain to detect subperiosteal/orbital abscess or sinus involvement.
5. Surgical Drainage: Indicated if there is no response to antibiotics in 24-48 hours, deteriorating vision, or a CT-confirmed orbital/subperiosteal abscess.

Q 4**SCENARIO**

A 38-year-old diabetic lady with known sinusitis develops a sudden, painful, forward protrusion of her right eye with ophthalmoplegia and fever. (a) What is the most likely diagnosis? (b) How do you differentiate it from Preseptal Cellulitis?

Model Answer

a) Diagnosis: Orbital Cellulitis (likely secondary to spread from paranasal sinuses, exacerbated by uncontrolled diabetes).

b) Differentiator Table:

Feature | Preseptal Cellulitis | Orbital Cellulitis

---|---|---

Site of Infection | Anterior to the orbital septum (lid tissues only) | Posterior to the orbital septum (orbital fat/muscles)

Proptosis | Absent | Present (globe pushed forward)

Ocular Motility | Normal (no pain on eye movements) | Restricted and painful (ophthalmoplegia)

Visual Acuity | Normal | May be decreased (due to optic nerve compression)

Pupil Reflexes | Normal | RAPD may be present (optic nerve dysfunction)

Systemic Symptoms | Mild | Severe fever, headache, nausea, leukocytosis

Q 5**DEFINITION****Define panophthalmitis and differentiate it from orbital cellulitis.****Model Answer**

Panophthalmitis is the acute purulent inflammation of all three coats of the eyeball (sclera, choroid, retina) and the intraocular cavities, which spills over into the Tenon's space and orbital tissues.

Differentiators:

1. Vision: Vision is completely lost (No Perception of Light, No PL) in panophthalmitis, whereas it is preserved or partially reduced in orbital cellulitis.
2. Ocular signs: Panophthalmitis has intraocular involvement (hypopyon, corneal cloudiness, yellow reflex from vitreous abscess/pus), while orbital cellulitis has a clear cornea, normal anterior chamber, and intact red reflex.
3. Pain: Pain is extremely severe and throbbing in panophthalmitis, worse than cellulitis.
4. Treatment: Panophthalmitis requires intravitreal antibiotics and often enucleation of the globe, whereas orbital cellulitis is treated with systemic antibiotics and drainage.

Q 6**ENUMERATE****Enumerate investigations for unilateral proptosis.****Model Answer**

1. Imaging: CT scan or MRI of the orbit and brain (best to show bone/fat details, muscle enlargement, and localized masses).
2. Ultrasonography: A-scan and B-scan of the orbit.
3. Endocrine: Thyroid Function Tests (T3, T4, TSH) and thyroid autoantibodies.
4. Vascular: Color Doppler imaging or Digital Subtraction Angiography (DSA - if CCF is suspected).
5. Biopsy: Fine needle aspiration biopsy (FNAB) or incisional biopsy of orbital mass.
6. Systemic Workup: CBC, ESR, blood cultures (if inflammatory/infective).

Q 7**SURGICAL PRINCIPLE**

Describe the clinical approach and surgical priority in treating thyroid eye disease (TED) orbitopathy.

Model Answer**Clinical Approach:**

Assess activity using the Clinical Activity Score (CAS) and severity (mild, moderate, severe/sight-threatening due to optic neuropathy).

1. Active phase: Immunosuppressants (IV Methylprednisolone) and orbital radiotherapy.
2. Inactive phase: Surgical rehabilitation.

Surgical Priority (Must be done in this strict chronological order):

1. Orbital Decompression: First priority, to expand the orbital cavity by removing bone/fat, relieving compressive optic neuropathy or severe proptosis.
2. Extraocular Muscle (Strabismus) Surgery: Second priority, to correct diplopia and realign the eyes after decompression.
3. Eyelid Surgery (Lid recession or blepharoplasty): Third priority, to correct eyelid retraction (Dalrymple's sign) and improve lid closure.

Q 8**DEFINITION**

Define exenteration and enumerate its indications.

Model Answer

Exenteration is the surgical removal of all the contents of the orbit, including the eyeball, extraocular muscles, orbital fat, connective tissue, and periosteum, and optionally the eyelids.

Indications:

1. Primary malignant orbital tumors (e.g., Rhabdomyosarcoma in children unresponsive to chemotherapy/radiotherapy, or Lacrimal gland adenocarcinoma).
2. Malignant tumors of the eyelids or conjunctiva that have invaded deep into the orbit (e.g., Sebaceous gland carcinoma, Malignant melanoma, Squamous cell carcinoma).
3. Extrabulbar extension of intraocular malignancies (e.g., Retinoblastoma or Choroidal melanoma with orbital invasion).
4. Aggressive, life-threatening fungal infections of the orbit (e.g., Rhino-orbital-cerebral Mucormycosis in diabetic/immunocompromised patients).

Q 9**SCENARIO**

A patient presents with pulsating proptosis, a bruit over the eye, and dilated episcleral veins. (a) What is the most probable diagnosis? (b) What is the surgical principle for treating this condition?

Model Answer

a) Diagnosis: Carotid-Cavernous Sinus Fistula (CCF) (abnormal communication between the carotid system and the cavernous sinus, leading to high-pressure arterial blood flowing retrogradely into orbital veins).

b) Surgical/Management Principle:

The principle is to close the abnormal fistula communication without compromising the main internal carotid artery flow, thereby lowering venous pressure and stopping the bruit/pulsation.

Method:

Interventional Endovascular embolization (done via angiography catheter inserting microcoils, Onyx glue, or detachable balloons into the cavernous sinus via the femoral vein/inferior petrosal sinus).